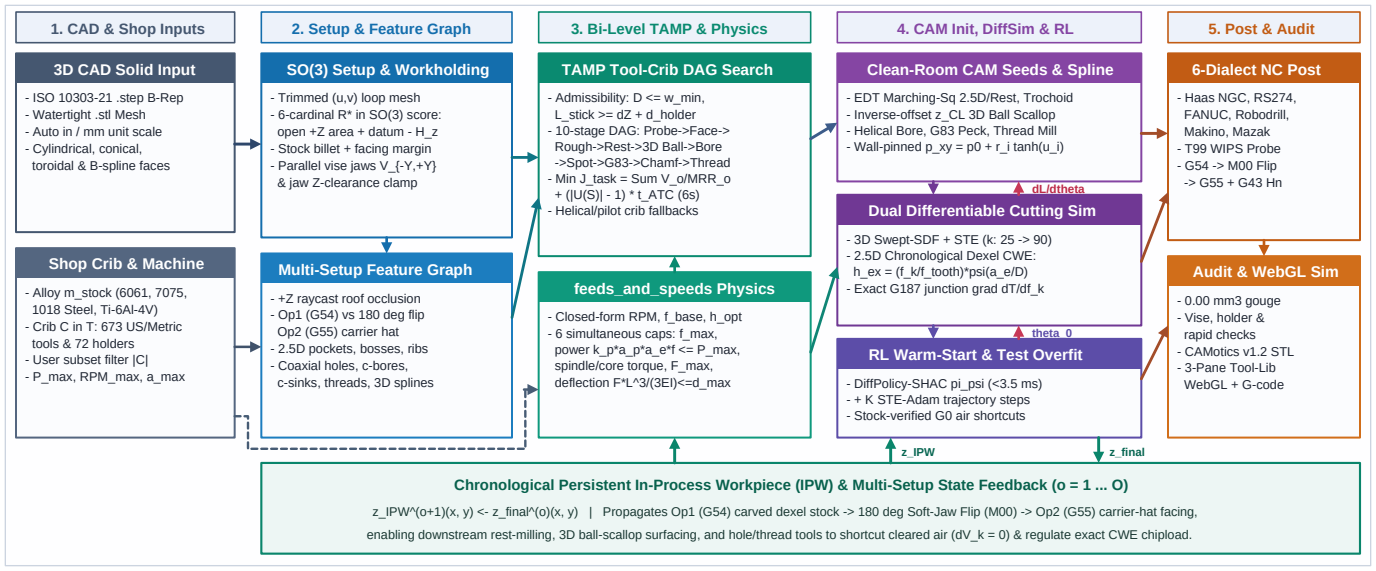
1. CAD & Shop Inputs
2. Setup & Feature Graph
3. Bi-Level TAMP & Physics
4. CAM Init, DiffSim & RL
5. Post & Audit
3D CAD Solid Input
- ISO 10303-21 .step B-Rep
- Watertight .stl Mesh
- Auto in / mm unit scale
- Cylindrical, conical, toroidal & B-spline faces
Shop Crib & Machine
- Alloy m_stock (6061, 7075, 1018 Steel, Ti-6Al-4V)
- Crib C in T: 673 US/Metric tools & 72 holders
- User subset filter |C|
- P_max, RPM_max, a_max
SO(3) Setup & Workholding
- Trimmed (u,v) loop mesh
- 6-cardinal R* in SO(3) score: open +Z area + datum - H_z
- Stock billet + facing margin
- Parallel vise jaws V_{-Y,+Y} & jaw Z-clearance clamp
Multi-Setup Feature Graph
- +Z raycast roof occlusion
- Op1 (G54) vs 180 deg flip Op2 (G55) carrier hat
- 2.5D pockets, bosses, ribs
- Coaxial holes, c-bores, c-sinks, threads, 3D splines
TAMP Tool-Crib DAG Search
- Admissibility: D <= w_min, L_stick >= dZ + d_holder
- 10-stage DAG: Probe->Face-> Rough->Rest->3D Ball->Bore ->Spot->G83->Chamf->Thread
- Min J_task = Sum V_o/MRR_o + (|U(S)| - 1) * t_ATC (6s)
- Helical/pilot crib fallbacks
feeds_and_speeds Physics
- Closed-form RPM, f_base, h_opt
- 6 simultaneous caps: f_max, power k_p*a_p*a_e*f <= P_max, spindle/core torque, F_max, deflection F*L^3/(3EI)<=d_max
Clean-Room CAM Seeds & Spline
- EDT Marching-Sq 2.5D/Rest, Trochoid
- Inverse-offset z_CL 3D Ball Scallop
- Helical Bore, G83 Peck, Thread Mill
- Wall-pinned p_xy = p0 + r_i tanh(u_i)
dL/dtheta
Dual Differentiable Cutting Sim
- 3D Swept-SDF + STE (k: 25 -> 90)
- 2.5D Chronological Dexel CWE: h_ex = (f_k/f_tooth)*psi(a_e/D)
- Exact G187 junction grad dT/df_k
theta_0
RL Warm-Start & Test Overfit
- DiffPolicy-SHAC pi_psi (<3.5 ms)
- + K STE-Adam trajectory steps
- Stock-verified G0 air shortcuts
6-Dialect NC Post
- Haas NGC, RS274, FANUC, Robodrill, Makino, Mazak
- T99 WIPS Probe
- G54 -> M00 Flip -> G55 + G43 Hn
Audit & WebGL Sim
- 0.00 mm3 gouge
- Vise, holder & rapid checks
- CAMotics v1.2 STL
- 3-Pane Tool-Lib WebGL + G-code
z_IPW
z_final
Chronological Persistent In-Process Workpiece (IPW) & Multi-Setup State Feedback (o = 1 ... O)
z_IPW^(o+1)(x, y) <- z_final^(o)(x, y) | Propagates Op1 (G54) carved dexel stock -> 180 deg Soft-Jaw Flip (M00) -> Op2 (G55) carrier-hat facing, enabling downstream rest-milling, 3D ball-scallop surfacing, and hole/thread tools to shortcut cleared air (dV_k = 0) & regulate exact CWE chipload.